

JEE MAIN REVISION MASTER: MATHEMATICAL REASONING 2026 1 QUESTIONS 1.67% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Mathematical Reasoning - Core Application	9 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Mathematical Reasoning - Core Application (High Yield — 9 PYQs)

Core Principle: Key problem pattern in Kcet Mathematical Reasoning.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The solution of $3(x-1) \leq 2(x-3)$ is..." [Ans: A]
- [2024 Shift] Q#2 (MCQ): "The negation of the statement "For every real number x ; x^2+5 is positive" is..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "The contrapositive of the statement. "If two lines do not intersect in the same plane, then they are parallel." is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The solution of $3(x-1) \leq 2(x-3)$ is

- (A) $x \leq -3$ **[Correct]**
- (B) $x \geq -3$
- (C) $x \leq 3$
- (D) $x \geq 3$

Correct Answer: A

Solution Analysis: Given inequality: $3(x-1) \leq 2(x-3)$ Open the brackets: $3x-3 \leq 2x-6$ Now subtract $2x$ from both sides: $x-3 \leq -6$ Add 3 to both sides: $x \leq -3$ So, the solution is: $x \leq -3$ Hence, the correct option is Option A .